AP Precalculus — Unit 2, Lesson 2.3

Exit Ticket — Answer Key

Name: ____________________________ Date: ________________ Period: _________

1. For $f(x) = 3 \cdot (0.5)^x$, identify each of the following.

   Initial value a: 3   Base b: 0.5   Growth or decay? **Decay**

2. State the end behavior of $g(x) = 5 \cdot 2^x$ using limit notation.

   $\lim_{x \to \infty} 5 \cdot 2^x = \infty$ $\lim_{x \to -\infty} 5 \cdot 2^x = 0$

3. A function has the following values.

x	0	1	2	3
$f(x)$	6	18	54	162

   Write $f(x)$ in the form ab^x. $f(x) = 6 \cdot 3^x$

   (Ratios: $18/6 = 3$, $54/18 = 3$, $162/54 = 3$; so $b = 3$ and $a = f(0) = 6$.)